

I am a second-year student at ENSIMAG, currently in the second year of the engineering program (Master’s level), specializing in Computer Science, Applied Mathematics and Finance. I enjoy working on low-level programming, mathematics, and computation, with particular interests in quantitative finance, artificial intelligence and computer graphics.

EDUCATION

Engineering Degree <i>ENSIMAG</i>	2026-2028
Track: Fundamentals and Innovative Technologies for Finance	
Double Bachelor’s Degree (Mathematics - Computer Science) <i>Université de Toulouse</i>	2023-2026
French General Baccalaureate	2023
Graduated with <i>Très Bien</i> honors, European English Section	

UNIVERSITY PROJECTS

Fourier Transforms Applied to Image Compression	2025-2026
- Toulouse Institute of Mathematics (supervisor: <i>Philippe Monnier</i>) - Study of Fourier transforms, image spectra, and the Cooley-Tukey FFT algorithm - Python implementation of FFT and DCT algorithms for JPEG-inspired image compression	
Motion Tracking Implementation	2025-2026
- Hand motion detection during piano playing using MMPose and MediaPipe - Object detection, trajectory analysis, and post-processing trajectory prediction	
Data Challenge Organized by ENEDIS	2025
- Development of algorithms for missing data imputation in time series - Data completion using regression models (XGBoost) and neural networks (DAE, LSTM) 24% reduction in MAE compared with the benchmark (linear interpolation)	
Study of Curve Length	2024-2025
- Toulouse Institute of Mathematics (supervisor: <i>Patrice Lassère</i>) - Theoretical study of the length of parametrized and regular curves, ellipses, and geodesics	

PERSONAL PROJECTS

Study and Implementation of the Cox-Ross-Rubinstein Model	Ongoing
- Study of no-arbitrage, replication, and risk-neutral pricing - Pricing of European and American options using backward induction - Design and implementation in C++23	
Video Game Development and Computer Graphics	Since 2018
- Data structures and decision-making AI algorithms (State Machines, Behavior Trees) - Advanced object-oriented programming (Java, GDScript, UE4) and 3D modeling with Blender - Implementation of a ray tracer in C++20	

SKILLS

Computer Science: Software Development, AI and Machine Learning (PyTorch, TensorFlow, Scikit-learn)

Mathematics: Probability, Statistics, Real Analysis, Linear Algebra, Mathematical Modeling

Programming Languages: Python, C++23, C, Java, OCaml, SQL; basic knowledge of R and Rocq

LANGUAGES

French : Native

English : B2 - Cambridge English Certificate (2023)

Spanish : B2